Department of Computer Science and Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. CSE 'A'
Course Code & Title: CS3391 Object Oriented Programming
Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Abstract Class**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen: Think Pair Share**
- **Justification:**

A cooperative, active learning approach where students initially work individually (Think), then collaborate in pairs or groups (Pair), and lastly present their idea to the class (Share) for the question posed by the instructor. The concept of abstraction is an essential for developing an application. Hence the students have to think on their own the ways of providing abstraction in an application.

- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
    - The instructor explained the concept of abstract class in the classroom for the duration of first 30 minutes. After that the Think- Pair- Share activity was conducted at the end of the class.
    - The instructor asked the students to think about the real time examples for providing abstraction.
    - Then the students are made as pair and they are sharing their responses with other partner and it is shown in Fig.1.
    - Finally, the students from two to three pairs shared their responses to the entire class so that all of them get different examples of using abstract class.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	1	2	2	1	3	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO9	PO10	PSO1	PSO2	PSO3
	3	2	1	1	3	1	1

Justification for correlation	The students apply the knowledge of abstract class	The students analyze and identify the real time example.	The students coordinate and demonstrate the activity as a team.	The students communicate the solution to others.	The concept is used in developing the software	The concept is used in security applications	The concept is used in solving problems in AI

- **Images / Screenshot of the practice:**

Fig 1: Students Involvement

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Different real time examples were identified by different teams.
    - When they collaborate in pairs, they can exchange knowledge and develop a deeper comprehension of the subject.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - Students' oral communication abilities are developed, and peer idea sharing is encouraged.
    - Speaking with their peers on this topic increases their expertise and self-assurance.
  - ❖ ***Challenges faced in implementation:***
    - When the question about the new real-time system was asked, the students found the task to be challenging and were unable to finish it in time.
    - More time is required for the discussion to make all students to actively participate.

References:

- ❖ https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5.Think_Pair_Share_DBMS_KVS.pdf
- ❖ https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share